

Q Conditional Prob.

Consider two events A & B

$$P(A) = 0.6$$

$$P(B) = 0.4$$

$$P(A \cap B)$$

$$\Rightarrow 0.2$$

$$P(A/B) \Rightarrow \frac{P(A \cap B)}{P(B)}$$

$$\Rightarrow \frac{0.2}{0.4} \Rightarrow 0.5$$

Q Cal. Goodness of fit ?

$$R^2 = ?$$

Actual value

$$y = [3, 2.5, 4, 5.5, 6.0]$$

Predicted
value $\hat{y}$

$$\Rightarrow [2.8, 2.7, 2.9, 5.6, 6.2]$$

$$\text{Mean } (\bar{y}) \Rightarrow \frac{3 + 2.5 + 4 + 5.5 + 6.0}{5}$$

$$\Rightarrow 4.2$$

	M	T	W	T	F	S	S
Sum Sq. total							

$$SS_T \Rightarrow \sum_{i=1}^n (y_i - \bar{y})^2$$

$$\Rightarrow (3 - 4.2)^2 + (2.5 - 4.2)^2 + (4 - 4.2)^2 + (5.5 - 4.2)^2 + (6 - 4.2)^2$$

$$\Rightarrow 9.3$$

Cal. SSE (Sum Sq. Error)

$$\Rightarrow \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$\Rightarrow (3 - 2.8)^2 + (2.5 - 2.7)^2 + \dots + (6 - 6.2)^2$$

$$\Rightarrow 0.14$$

$$R^2 \Rightarrow 1 - \frac{SSE}{SST}$$

$$\Rightarrow 1 - \frac{0.14}{9.3} \Rightarrow 0.9849$$

K-mode clustering "Hamming distance"

$$\text{distance}(x_i, y_i) = \sum_{j=1}^m S(x_{ij}, y_{ij})$$

$$S(x_{ij}, y_{ij}) = \begin{cases} 0 & \text{if } x_{ij} = y_{ij} \\ 1 & \text{if } x_{ij} \neq y_{ij} \end{cases}$$

Q 4 data points ; 2 features.
 $k = 2$.

A	B
1	2
1	3
0	3
0	2

initial mode for
clusters $(1, 2)$ & $(0, 3)$
 m_1 m_2

sol

Data point	Similarity		cluster Assignment
	m_1	m_2	
(1, 2)	0	2	C_1
(1, 3)	1	1	C_1 or C_2
(0, 3)	2	0	C_2
(0, 2)	1	1	C_1 or C_2

Q K-medoid

Manhattan distance

$$|x_1 - x_2| + |y_1 - y_2|$$

Q

data point

X	Y
1	2
1	3
0	3
0	2

$K = 2$

initial medoids

$m_1 (1, 2)$

$m_2 (0, 3)$

Sol

data point	Dist. to m_1	Dist. to m_2	Clusters Assignment
(1, 2)	0	2	C_1
(1, 3)	$ 1-1 + 2-3 = 1$	1	C_1 or C_2
(0, 3)	$ 1-0 + 3-2 = 2$	0	C_2
(0, 2)	$ 1-0 + 2-2 = 1$	2	C_1 or C_2

total cost $\Rightarrow 0 + 1 + 1 + 0 = 2$

~~Recall~~ assign new medoids
 Q Calc the distances
 Q total cost

Smooth $\rightarrow 0$
Rough $\rightarrow 1$

Naive Bayes

blight	texture	Label
150	0	Apple
160	0	Apple
170	1	orange
180	1	orange

new i/p to classify (165, 0)

Sol Step ①

$$P(\text{Apple}) \Rightarrow 2/4 \Rightarrow 0.50$$

$$P(\text{orange}) \Rightarrow 2/4 \Rightarrow 0.50$$

Step ② Post Prob. for
Categorical feature ("weight")

$$\text{Mean of Apple } \mu_{\text{apple}} \Rightarrow \frac{150+160}{2}$$

$$\Rightarrow 155$$

$$\text{Std. deviation}_{\text{Apple}} \Rightarrow \sqrt{\frac{(150-155)^2 + (160-155)^2}{2}}$$

$$\Rightarrow \sqrt{\frac{25+25}{2}} \Rightarrow 5$$

$$\mu_{orange} \Rightarrow 175$$

$$\sigma_{orange} = \sqrt{25} = 5$$

Step 3. make predictions

$$P(\text{Apple} | 165, 0) \quad P(165/\text{Apple})$$

$$P(0/\text{Apple})$$

(Gaussian distribution) $P(\text{Apple})$

$$\Rightarrow P(165/\text{Apple}) \Rightarrow \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$\Rightarrow \frac{1}{\sqrt{2\pi \times 25}} e^{-\frac{(165-175)^2}{2(5)^2}}$$

$$\Rightarrow 0.048$$

$$\Rightarrow P(0/\text{Apple}) \Rightarrow 2/2 \Rightarrow 1$$

So by ①

$$P(\text{Apple} | 165, 0) \Rightarrow 0.048 \times 1 \times 0.50$$

$$\Rightarrow 0.024 \quad \text{HQ}$$

Similarly

$$P(\text{orange} | (165, 0))$$

$$\propto P(165 | \text{orange}) \cdot P(0 | \text{orange})$$
$$P(\text{orange})$$

$$\propto 0.040 \times 0 \times 0.80$$

$$\propto 0$$

So fruit is 'Apple'.

Q

Maine Bayls.Weather

Sunny

Sunny

Rainy

Rainy

Rainy

Sunny

Play tennis

No

No

Yes

Yes

No

Yes

new
i/pWeather | Rainy

then o/p?

one i/p & one o/p

Sol

Step ① Cal. Prior Probabilities

$$P(\text{Yes}) \Rightarrow \frac{3}{6} \Rightarrow 0.50$$

$$P(\text{No}) \Rightarrow \frac{3}{6} \Rightarrow 0.50$$

Step ②

Cal. Posterior probs

$$P(\text{Rainy} / \text{Yes}) \Rightarrow \frac{2}{3} \Rightarrow 0.666$$

$$P(\text{Rainy} / \text{No}) \Rightarrow \frac{1}{3} \Rightarrow 0.33$$

$$P(\text{Sunny} / \text{yes}) \Rightarrow \frac{1}{3} \Rightarrow 0.33$$

$$P(\text{Sunny} / \text{no}) \Rightarrow \frac{2}{3} \Rightarrow 0.66$$

Step 3 Cal. Posterior Prob using
Bayes theorem
(naive Prediction)

~~P~~

$$P(\text{yes} / \text{Rainy}) \Rightarrow \frac{P(\text{Rainy} / \text{yes}) \cdot P(\text{yes})}{P(\text{Rainy})}$$

$$\alpha = P(\text{Rainy} / \text{yes}) P(\text{yes})$$

$$\alpha = 0.66 \times 0.50$$

$$\alpha = 0.332$$

$$P(\text{no} / \text{Rainy}) \Rightarrow \frac{P(\text{Rainy} / \text{no}) P(\text{no})}{P(\text{Rainy})}$$

$$\alpha = P(\text{Rainy} / \text{no}) P(\text{no})$$

$$\alpha = 0.33 \times 0.50$$

$$\alpha = 0.166$$

Step 1 is "yes"

Q1 K-nn. $K=3$ Given

	x (i/p)	y (o/p)	
S.No.	x_1	x_2	Class
1	1	2	0
2	2	3	0
3	3	3	1
4	5	2	1
5	6	4	1

New point to classify (4,3)

Sol Calc distance of new point with each point

S.No	x_1	x_2	dist. from (4,3)	
1	1	2	3.16	
2	2	3	2	} min distance
3	3	3	1	
4	5	2	1.41	
5	6	4	2.24	

Class (0, 1, 1)

majority is 1

So class of (4,3) is 1

SVM

$$wx + b \geq 1 \text{ (class +1)}$$
$$wx + b \leq -1 \text{ (class -1)}$$

Q.

	x	class	
x_1	1	-1	y_1
x_2	2	+1	y_2

Decision boundary ?
classify new point $x = 4$

sol

decision function

$$y = w \cdot x + b$$

for i/p $x_1 = 1$; $y_1 = -1$ or

$$y_1 = w(1) + b$$

$$w(1) + b \leq -1 \quad \text{--- (1)}$$
$$w + b \leq -1$$

for $x_2 = 2$; $y_2 = +1$

$$w(x) + b \geq 1$$

$$2w + b \geq 1 \quad \text{--- (2)}$$

to solve this eq. (1) $w + b + 1 \leq 0$

eq. (2) $2w + b - 1 \geq 0$

$$\text{eq. ①} = \text{eq. ②}$$

$$w + b + 1 = 2w + b - 1$$

$$w - 2w = -1 - 1$$

$$-w = -2$$

$$\boxed{w = 2}$$

$$\text{eq. ①} \quad w + b + 1 = 0$$

$$2 + b + 1 = 0$$

$$2 + b + 1 = 0$$

$$b = -3$$

$$\boxed{b = -3}$$

decision function

$$\boxed{y = 2x - 3}$$

for new i/p $x = 4$

$$y = 2 \times 4 - 3$$

$$= 5$$

$$\boxed{\text{class} + 1} \quad \text{for } x = 4$$